Jorge Manuel Torre

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Education

Virginia Tech Blacksburg, VA
B.S. Computer Science, GPA: 3.37/4.0, May 2026

Skills

Languages: Python, TypeScript, Rust, C, SQL, Bash, Lua
Web & Frameworks: React, Three.js, Vite, Django, REST APIs
Infrastructure: Linux, Docker, Proxmox, OPNsense, NixOS, Nginx, PostgreSQL, Git, CI/CD, SSH

Experience

Systems Administrator / Infrastructure Engineer January 2019 – December 2023
Independent Systems Administration *Proxmox, Linux, NixOS, Docker, OPNsense*

- Administered and maintained a Linux infrastructure environment spanning Proxmox virtual machines, NixOS systems, and Docker-based services.
- Configured and managed network infrastructure using OPNsense, including firewall rules, port forwarding, DNS, routing, and service accessibility.
- Develop custom scripts, plugins, and server-side software to automate administrative tasks, service management, and system maintenance.
- Managed remote administration through SSH, access controls, service configuration, and secure network practices.
- Deployed and maintain containerized services using Docker, including networking, persistent storage, configuration, and service dependencies.
- Troubleshooted system, application, and network issues using logs, resource utilization, configuration analysis, and command-line tooling.
- Performed software upgrades, configuration changes, backups, and recovery procedures while maintaining service availability.

Projects

Garage64: Custom Entity Model Studio • GitHub August 2025 – Present
React, Three.js, Django, SQLite, Docker

- Developing a self-hosted web application for creating custom Minecraft vehicle models and texture packs using the OptiFine Custom Entity Model (CEM) format.
- Building an interactive React and Three.js 3D editor that supports real-time vehicle customization through modular part swapping and texture painting.
- Implementing a Django REST API for managing entity metadata, asset configurations, and project persistence.
- Creating an automated export pipeline for generating `.jem` and `.jpm` asset files from user-created models.
- Containerizing the application with Docker Compose to streamline local development and deployment.

PokedexHex: Terminal Pokédex & Pokémon Bank • GitHub October 2026 – Present
Rust, ratatui, crossterm

- Porting PKHeX.Core from C# to Rust (35K+ lines) to read and write save files for every mainline Pokémon game from Red/Blue to Legends: Z-A, including per-generation encryption and string encoding.
- Building a ratatui terminal app that backs up every Pokémon found in a folder of saves into a deduplicated bank and tracks species, form, shiny, and living-dex completion across games.
- Added a PKHeX-style save editor for party, box, and trainer data, backed by round-trip tests for each generation's save and text formats.

CHUD: Catastrophic Harmful Update Detection • GitHub January 2026 – April 2026
Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth *Team of 4 (CS Capstone)*

- Tested the LoX defense (Perin et al., COLM 2025) against an iterative LoRA fine-tuning attack on Llama-2-7b using benign GSM8k and adversarial BeaverTails samples.
- Built an Attack Success Rate (ASR) evaluation pipeline with Llama-Guard-3-8b as the judge model and ran 30+ experiments across attack mixes and learning rates.
- Found that LoX lowered ASR from 91–96% to 37–63% under direct harmful fine-tuning, but an iterative mixed-data attack raised ASR back to 89%.
- Set up x86-64 and aarch64 VT GPU cluster environments for the team and ran QLoRA fine-tuning evaluations for the final analysis.

Leadership & Community

Team Lead, Database Management Systems May 2025 – August 2025

- Led a 3-person team through schema design and SQL development, then integrated the team's work into the final submission.

Team Lead, Intermediate Software Design January 2025 – April 2025

- Led a 5-person team through requirements gathering, sprint planning, code reviews, and final project delivery.

Garage32 (Community Project) March 2020 – December 2021

- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.